

Grade 9 (Form 3) ICT – Exam Revision & Predicted Questions

Based on the MIE Grade 9 ICT units: Computer Fundamentals & Operations · Word Processing · Spreadsheet · Presentation · Internet · Multimedia · Health, Safety & Ethics. Programming and Database (Grade 9+ Extended units) are included as bonus sections.

Note: the book itself could not be opened when this was prepared, so notes follow the standard Grade 9 syllabus for each unit. Check each point against your own copy of the book and your teacher's notes.

Unit 1: Computer Fundamentals and Operations

- **Data vs information; computer system** = hardware + software + people + data. Cycle: input → process → output → storage.
- **Number systems:** binary (0,1), decimal. Convert binary to decimal using place values 128, 64, 32, 16, 8, 4, 2, 1. Example: 1101 = 8+4+0+1 = 13. Decimal 10 = 1010.
- **Storage units:** bit, byte (8 bits), KB, MB, GB, TB (×1024 each step). Text uses ASCII/Unicode codes.
- **CPU:** ALU (calculations/logic), control unit, registers; clock speed in GHz; cores; **cache** memory. **RAM** (volatile) vs **ROM** (non-volatile); **virtual memory**.
- **Storage:** magnetic (HDD), optical (CD/DVD/Blu-ray), solid-state (SSD, USB, SD card), cloud. Compare capacity, speed, portability, cost.
- **Software:** operating system (manages memory, files, devices, users, multitasking), utilities (antivirus, backup, disk defragmenter, compression), application software; **device drivers**; **firmware**; open-source vs proprietary; licences.
- **Networks:** LAN, WAN, WLAN; client-server and peer-to-peer; topologies star, bus, ring (star: easy to add devices, failure of one device does not affect others); devices: router, switch, hub, modem, NIC, access point; wired vs wireless.
- **Emerging technologies:** AI, IoT, cloud computing, robotics, VR/AR, 3D printing, blockchain basics.
- **Operations:** managing files/folders, shortcuts, searching, compressing, backup and restore, installing/uninstalling software, updating, troubleshooting basics (restart, check cables, safe mode).

Unit 2: Word Processing

- **Formatting:** fonts, paragraph spacing, indents, tabs, bullets/numbering, borders and shading.
- **Styles and table of contents;** headers, footers, page numbers, section/page breaks, columns, watermark.
- **Graphics and objects:** pictures with text wrapping, shapes, text boxes, SmartArt, charts, drop caps, captions, footnotes/endnotes, hyperlinks, bookmarks.
- **Tables:** merge/split, align, sort, simple formulas, convert text to table.
- **Mail merge:** main document + data source + merge fields → personalised letters, labels, envelopes.
- **Collaboration:** track changes, comments, compare documents, protect with password, share via cloud.
- **Referencing:** acknowledge sources; bibliography.
- **Document design:** choose the right format (letter, newsletter, report, brochure), consistent layout, readable fonts, proofread (spelling ≠ correct meaning).

Unit 3: Spreadsheet

- **Cell references:** relative, absolute (\$A\$1), mixed; reference to another worksheet (Sheet2!A1).
- **Functions:** SUM, AVERAGE, MAX, MIN, COUNT, COUNTA, ROUND, IF, nested IF, COUNTIF, SUMIF, VLOOKUP basics. IF example: =IF(A1>=50,"Pass","Fail"). Nested: =IF(A1>=70,"A",IF(A1>=50,"B","C")).
- **BODMAS** in formulas; percentage: =B2/C2*100 or format as %.
- **Formatting:** number formats, conditional formatting (highlight cells by rule), merge, wrap text, protect cells.
- **Charts:** column, bar, line, pie, scatter; choose by purpose; title, axis labels, legend, data labels.
- **Data tools:** multi-level sort, filter, freeze panes, data validation (drop-down lists), find & replace, what-if analysis.
- **Errors:** #DIV/0!, #NAME?, #VALUE!, #REF!, #####.
- **Planning a spreadsheet model:** label inputs, use formulas not typed numbers, test with sample data, document assumptions. Examples: budget, marks sheet, payroll, sales.

Unit 4: Presentation

- **Planning:** objective, audience, storyboard, content outline.
- **Slide master and layouts** for consistency; templates/themes.
- **Content:** text, images, charts, tables, audio, video, SmartArt, hyperlinks, action buttons.
- **Animation vs transition;** timing; rehearse timings; self-running show (kiosk).
- **Principles:** simplicity, contrast, alignment, repetition, limited fonts (max 2) and colours, 6×6 rule (≤6 words per line, ≤6 lines).
- **Delivery:** speaker notes, handouts, projector setup, eye contact, voice, answering questions.
- **Evaluate** a presentation: content accuracy, design, audience engagement, timing.
- **Share/export:** .pptx, .ppsx, PDF, video, cloud sharing.

Unit 5: Internet

- **Internet vs WWW; protocols:** HTTP/HTTPS (web), SMTP/POP/IMAP (email), FTP (files), TCP/IP; **IP address, DNS, URL, ISP, bandwidth.**
- **Web 2.0 services:** social media, blogs, wikis, video sharing, cloud collaboration, online forms.
- **Searching:** advanced search, Boolean operators, filters; evaluating reliability (author, date, bias, domain, cross-check).
- **E-services:** e-commerce, e-banking, e-government, e-learning, telecommuting, online booking.
- **Email:** CC/BCC, attachments, signature, rules, netiquette; spam, phishing, spoofing.
- **Security:** firewall, antivirus, encryption, HTTPS/SSL padlock, two-factor authentication, strong passwords, backups, updates, safe public Wi-Fi use; cookies and privacy.
- **Cloud computing:** store files online; advantages: access anywhere, share, backup; disadvantages: need Internet, privacy concerns.
- **Impact of the Internet:** positive (access, communication, jobs) and negative (misinformation, cybercrime, addiction, digital divide).

Unit 6: Multimedia

- **Multimedia:** text, graphics, audio, video, animation; interactive vs linear.

- **Graphics:** bitmap vs vector; resolution (dpi), colour depth, RGB vs CMYK; formats JPG, PNG, GIF, SVG; editing tools: crop, resize, layers, filters, retouch.
- **Audio:** sampling rate and file size; MP3 (lossy), WAV (uncompressed); recording and editing: trim, fade, volume, noise removal.
- **Video:** frame rate (fps), resolution (HD, 4K), MP4; editing: trim, split, transitions, titles, audio track, export.
- **Compression:** lossy vs lossless; reduces file size for storage and faster streaming/downloading.
- **Animation:** frame-by-frame, keyframes, tweening; GIF animation.
- **Multimedia project steps:** plan → storyboard → collect/create media → edit → assemble → test → publish → evaluate.
- **Copyright and licences:** Creative Commons, royalty-free; attribute sources.

Unit 7: Health, Safety and Ethics

- **Health:** eye strain, RSI, back/neck pain, headache, stress, addiction, inactivity. **Ergonomics:** chair, desk, screen position, lighting, breaks.
- **Safety:** electrical and fire safety, cable management; disposal of **e-waste**; **green ICT:** power saving, recycling, paperless office.
- **Cyber threats:** virus, worm, trojan, spyware, ransomware, phishing, hacking, identity theft, denial of service, social engineering. Protection: layers of security.
- **Cybercrime and cyberbullying:** report to police/adult, keep evidence, block.
- **Ethics:** copyright, plagiarism, piracy, fair use, netiquette, honesty, respect for privacy, responsible social media use, fake news.
- **Laws:** Data Protection Act (personal data must be accurate, secure, used lawfully), Computer Misuse and Cybercrime Act (unauthorised access, hacking, illegal), Copyright Act.
- **Digital footprint & digital citizenship:** be safe, responsible, respectful.
- **Impact of ICT on society:** jobs, automation, communication, education, health care; digital divide.

Bonus: Problem Solving, Programming and Database (Grade 9+ Extended)

- **Problem solving:** define → design algorithm → code → test → debug → evaluate. Flowchart symbols: oval, rectangle, parallelogram, diamond.
- **Programming:** sequence, selection, loops, variables, data types (integer, real, string, Boolean), operators; trace tables; syntax/logic/runtime errors; Scratch or Python-style pseudocode.
- **Database:** table, record, field, primary key, data types; objects: table, query, form, report; advantages over paper filing; data validation.
- **Query example:** Age > 15 AND Town = "Quatre Bornes".

Key Definitions – Quick Glossary

Term	Meaning
Bit / Byte	Binary digit / 8 bits
Cache	Small fast memory near the CPU
Virtual memory	Hard-disk space used as extra RAM
Device driver	Software that lets the OS communicate with a device
Firmware	Software stored permanently on a device chip
IP address	Unique number identifying a device on a network
DNS	Translates domain names to IP addresses
HTTPS	Secure version of HTTP using encryption
Two-factor authentication	Login using two different proofs, e.g. password + code
Conditional formatting	Cell formatting that changes automatically by a rule
Lossy compression	Smaller file, some quality lost
Encryption	Scrambling data so only authorised users can read it
Digital divide	Gap between those with and without ICT access
Phishing	Fake messages designed to steal information
Digital footprint	Traces left by online activity

Predicted Questions

Section A – Multiple Choice (1 mark each)

1. Binary 1101 equals decimal:
A 11 B 12 C 13 D 14
2. 1 GB is approximately:
A 1024 KB B 1024 MB C 1024 TB D 8 bits
3. Which memory is volatile?
A ROM B RAM C SSD D DVD
4. Software that lets the OS control a printer is a:
A Utility B Driver C Browser D Compiler
5. In which topology does the failure of one device not stop the others?
A Bus B Ring C Star D None
6. Which protocol is used to send email?
A FTP B SMTP C HTTP D DNS
7. The padlock and https in the address bar show that:
A The site is free B Data is encrypted C The site is fast D It is a school site
8. =IF(A1>=50,"Pass","Fail") for A1 = 45 gives:
A Pass B Fail C 45 D Error
9. Which function counts cells that meet a condition?
A SUM B COUNTIF C ROUND D MAX
10. \$B\$2 is a(n):
A Relative reference B Absolute reference C Range D Function
11. Cell formatting that changes automatically by a rule is:
A Merge B Conditional formatting C Wrap text D Sort
12. A fake email asking for your bank details is:
A Spam only B Phishing C Cookie D Firewall
13. Which format uses lossy compression?
A PNG B ZIP C JPG D WAV
14. Which colour mode is used for printing?
A RGB B CMYK C HSL D BW
15. The law against hacking in Mauritius is the:
A Copyright Act B Computer Misuse and Cybercrime Act C Road Act D Fisheries Act
16. The gap between those who have and do not have ICT access is the:
A Digital footprint B Digital divide C Firewall D Bandwidth
17. In a database a primary key:
A Is repeated B Uniquely identifies a record C Is a report D Is a query

Section B – Short Answer

1. Convert 10110 (binary) to decimal and 25 (decimal) to binary. [4]

2. Differentiate between RAM and ROM, and between HDD and SSD. [6]
3. Explain the role of the operating system (four functions). [4]
4. Differentiate between LAN and WAN; give an advantage of a star topology. [4]
5. Define cache memory and virtual memory. [4]
6. Explain what a device driver and a utility program are. [4]
7. Explain how mail merge works and give two uses. [4]
8. State three ways of using styles and a table of contents to improve a long document. [3]
9. Differentiate between relative and absolute references with examples. [4]
10. Write formulas for: (a) average of B2:B20, (b) number of cells in C1:C30 greater than 50, (c) grade 'A' if $D2 \geq 70$, 'B' if ≥ 50 , else 'C'. [5]
11. Explain data validation in a spreadsheet and give an example. [3]
12. Choose the best chart for (a) monthly rainfall change, (b) share of budget, (c) comparing marks of pupils; justify. [6]
13. State the 6×6 rule and four other good slide design rules. [5]
14. Explain the terms IP address, DNS and ISP. [6]
15. Give four ways to protect yourself and your data online. [4]
16. Explain two advantages and two disadvantages of cloud storage. [4]
17. Differentiate between lossy and lossless compression. [3]
18. Explain frame rate and resolution in video. [4]
19. Describe five health problems from ICT use with solutions. [5]
20. Define phishing, ransomware, spyware and identity theft. [4]
21. State three principles of the Data Protection Act. [3]
22. Explain what is meant by digital footprint and how to manage it. [3]

Section C – Practical / Structured Questions

1. **Word processing:** Design a 2-page school newsletter. Describe the steps to set 2 columns, insert a picture with text wrapping, create a header and footer, add page numbers, and use mail merge to send it with personalised greetings. [10]
2. **Spreadsheet:** A shop records Item, Price, Quantity in A1:C10. (a) Write a formula for the total cost of row 2. (b) Total of all totals. (c) Add 15% VAT using an absolute reference to a VAT cell. (d) Use conditional formatting to highlight totals above Rs 1000. (e) Create a chart of Item vs Total and state its type. [10]
3. **Presentation:** Plan a 7-slide presentation titled 'Staying Safe Online' for Grade 7 pupils. Include the slide master, a chart, a hyperlink and rehearsed timings, and describe your delivery. [8]
4. **Network:** A school lab has 20 computers and 2 printers. (a) Suggest a network type and topology. (b) Name four devices needed. (c) State two benefits and two risks, with protection measures. [10]

5. Multimedia: You must make a 2-minute video about school sports day. (a) Describe the steps from planning to publishing. (b) Suggest file formats and compression. (c) Mention copyright considerations. [8]

6. Scenario: A friend shares your private photos without permission and a stranger asks you to meet. (a) Identify the issues. (b) Give four actions. (c) Which laws may apply? [8]

7. Bonus – database: A clinic stores Patient ID, Name, Age, Town. (a) Identify records, fields and the primary key. (b) Write a query criterion for patients aged over 60 from Curepipe. (c) Give two validation checks. [8]

8. Bonus – algorithm: Draw a flowchart or write pseudocode to input 5 marks, calculate the average and output 'Pass' if the average is at least 50. [8]

Answer Guide

Section A

1-C, 2-B, 3-B, 4-B, 5-C, 6-B, 7-B, 8-B, 9-B, 10-B, 11-B, 12-B, 13-C, 14-B, 15-B, 16-B, 17-B

Section B – Model answers

- 1. $10110 = 16+4+2 = 22$; $25 = 11001$.
- 2. RAM: temporary, volatile, working memory; ROM: permanent, start-up instructions. HDD: magnetic, cheaper, slower, moving parts; SSD: flash, faster, silent, more expensive.
- 3. Manages memory, files, hardware devices, multitasking, security/users, user interface.
- 4. LAN: small area; WAN: large area. Star: failure of one cable/device does not affect others; easy to add devices.
- 5. Cache: small, very fast memory close to the CPU storing frequently used data. Virtual memory: disk space used as extra RAM.
- 6. Driver: lets OS communicate with hardware. Utility: maintenance software, e.g. antivirus, backup.
- 7. A main document with merge fields is combined with a data source to create personalised copies; e.g. letters, labels.
- 8. Consistent formatting, quick navigation, automatic table of contents that updates.
- 9. Relative changes when copied ($B2 \rightarrow B3$); absolute stays fixed ($\$B\2).
- 10. (a) $=\text{AVERAGE}(B2:B20)$ (b) $=\text{COUNTIF}(C1:C30, ">50")$ (c) $=\text{IF}(D2 >= 70, "A", \text{IF}(D2 >= 50, "B", "C"))$.
- 11. Restricts entries to valid data, e.g. marks between 0 and 100, or a drop-down list.
- 12. (a) line (b) pie (c) column/bar.
- 13. ≤ 6 words per line and ≤ 6 lines; contrast, readable font, relevant images, consistent theme, limited animation.
- 14. IP address: device number; DNS: names to IPs; ISP: provides Internet access.
- 15. Strong unique passwords, two-factor authentication, antivirus/firewall, updates, avoid unknown links, HTTPS, privacy settings, backup.
- 16. Advantages: access anywhere, easy sharing/backup. Disadvantages: needs Internet, privacy and security concerns.
- 17. Lossy loses some data permanently; lossless can restore exactly.
- 18. Frame rate: frames per second; resolution: number of pixels (e.g. 1920×1080).
- 19. Eye strain – breaks; RSI – correct typing posture; back pain – good chair; headaches – lighting/brightness; stress or addiction – limit screen time.
- 20. Phishing: fake messages to steal data; ransomware: locks files for payment; spyware: secretly collects data; identity theft: using someone's details illegally.
- 21. Data must be processed lawfully, accurate, kept secure, kept no longer than necessary, used only for stated purpose.
- 22. Trace of data left online; manage with privacy settings, careful posting, deleting old accounts.

Section C – Hints

- Q2: Total =B2*C2; grand total =SUM(D2:D10); VAT =D2*\$F\$1; conditional formatting rule 'greater than 1000'; column chart.
- Q4: Star topology with switch, router, NIC, cables/access point; firewall, antivirus, passwords.
- Q6: Privacy violation, cyberbullying, stranger danger; report, block, keep evidence, tell an adult; Data Protection Act, Computer Misuse and Cybercrime Act.
- Q7: Record = row; field = column; key = Patient ID; criteria Age > 60 AND Town = "Curepipe"; range and presence checks.
- Q8: Input m1..m5; sum; avg = sum/5; if avg ≥ 50 output Pass else Fail.

Exam tips: Read each question twice · Match your answer length to the marks · Show formulas and working · Revise definitions · Practise flowcharts and spreadsheet functions · Manage your time.